Solution Architecture

Treavue — Personalized Travel Itinerary Planner

Project documentation prepared for portfolio and repository use

Layer 1 — Presentation

React/Vite/TypeScript pages collect preferences, display destinations and itineraries, and manage trip interactions.

Layer 2 — API

FastAPI exposes destination, itinerary, chat, and trip routes. Request/response schemas validate structured data.

Layer 3 — Logic

Itinerary generation selects and orders available activity schedules using user preferences and current project data.

Layer 4 — Persistence

Supabase PostgreSQL stores trip metadata and the complete nested trip object in JSONB. Frontend localStorage is a resilience fallback.

Data flow

Browser → FastAPI → itinerary logic or Supabase → JSON response → browser UI.

Security boundary

Do not publish the privileged Supabase secret. Before public deployment, verify Supabase Auth tokens, enforce owner checks in the backend, and configure RLS for per-user rows.

Future architecture improvements

Verified authentication and ownership, trusted live data providers, route APIs, observability, and automated end-to-end tests.